

CognitiveQG Annotation Guideline

Background:

Cognition refers to the mental processes people use to understand information, such as perceiving, remembering, interpreting, and reasoning. Facione's critical thinking framework organizes these processes into six skills: interpretation, analysis, evaluation, inference, explanation, and self-regulation. These skills describe how people make deliberate judgments when assessing arguments. Importantly, cognitive processes at each stage may vary depending on the downstream task (e.g., question generation) and educational level. In this work, we investigate differences between human and LLM cognition in question generation.

- **Dimensional Framework:** Facione's theory establishes that critical thinking comprises essential dimensions: cognitive skills (interpretation, analysis, evaluation, inference, explanation, and self-regulation) that represent the ability to perform critical thinking tasks
- **Skills-Disposition Interdependence:** The Delphi consensus framework demonstrates that effective critical thinking requires cognitive competency, as possessing critical thinking skills does not guarantee their application—individuals must also be disposed to use these skills consistently across different contexts and problems

Research Question:

"How does Facione's critical thinking framework align with LLM cognition in question generation, and where does it misalign across cognitive phases and question types?"

"After identifying where misalignment occurs, to what extent does alignment with human cognition result in improvement in Socratic and question generation abilities?"

Objective:

Create the best annotation scheme and task setting that best: 1) reflects human and model cognitions, 2) allows easy annotation and faithful evaluation, and 3) provides a deeper and structured analysis of both cognition

Based Dataset:

[FOCUS2025](#)

FOCUS dataset Component:

- Argument + Weakness Type + Related Weakness Span + FOCUS Question.

CognitiveQ Dataset

- Input: Argument

Annotation Scheme:

- Facione Theory: Interpretation, Analysis, Inference, Evaluation, Explanation, Self-Regulation
- Also, after going through the Facione Theory annotation, please create up to 3 questions (≤ 3 questions) based on critical thinking.
- Annotation Motivation: [Critical Thinking: What It Is and Why It Counts](#)

Annotation Resource:

1. **Unzip** the zip file.
2. Open **Terminal** in that folder
3. Upload the csv file and start the annotation

Annotation Process:

Read the argument paired with the FOCUS question, then complete the annotation scheme for interpretation, analysis, inference, evaluation, explanation and self regulation.

- Key Points:
- **Interpretation:** Use your first impression to capture natural human thinking
- **Multiple spans:** If the argument has multiple correct spans, choose the first one (Please highlight the necessary component; the shorter the span, the better)
- It is not necessary to answer every scheme, as sometime human also cannot process
- **Completion:** click "Save Progress (JSON)" and "Save Progress (CSV)" and submit both files

1. Interpretation

Facione Theory: To comprehend and express the meaning and significance of a wide variety of experiences, situations, datasets, events, judgments, conventions, beliefs, rules, procedures, or criteria.

1. What does this mean?
2. What's happening?
3. How should we understand that (e.g., what he or she just said)?
4. What is the best way to characterize/categorize/classify this?
5. In this context, what was intended by saying/doing that?
6. How can we make sense out of this (experience, feeling, statement)?

Annotation Scheme and Task with Example:

Argument:

From elementary school all the way to now, I have always done poor academically. I was always wondering if I have an intellectual disorder since I just couldn't grasp concepts as well as the other students.

Initial Understanding of Target:

- Restating the core target of the argument (Topic, Person, Idea)
- The argument talks about _____.
- Eg: The argument talks about intellectual disorder

Stance Toward the Core Target:

- Dropdown: [favor, against, neutral]
- *Select whether the author supports, opposes, or remains neutral about the main topic.*
- Eg: [against] intellectual disorder.

Knowledge Domain to Understand:

- Dropdown: [Technical/Scientific, Legal/Policy, Cultural/Social, Economic/Business, Government/Law, No specialized knowledge]
- Choose the type of background knowledge needed to fully understand this argument.
- *Choose what type of background knowledge is needed to fully understand this argument.*
- Eg: No specialised knowledge

Dropdown Explanations:

- [Knowledge Domain List] :
 - **Technical/Scientific:** Requires specialized knowledge in STEM fields (science, technology, engineering, mathematics), including research methods, technical processes, or scientific concepts.
 - **Legal/Policy:** Requires knowledge of laws, regulations, legal procedures, policy frameworks, or institutional governance structures.
 - **Cultural/Social:** Requires understanding of cultural norms, social dynamics, historical context, demographics, or community-specific knowledge.
 - **Economic/Business:** Requires knowledge of financial concepts, market dynamics, business operations, economic principles, or industry-specific practices.
 - **Government/Law:** Requires knowledge of governmental structures, political processes, administrative procedures, or how laws are implemented and enforced.
 - **No specialized knowledge:** Can be understood and annotated using general knowledge that an educated layperson would possess.

2. Analysis

Facione Theory: To identify the intended and actual inferential relationships among statements, questions, concept descriptions, or other forms of representation intended to express belief, judgment, experience, reasoning, or opinion

1. Please tell us again your reasons for making that claim.
2. What is your conclusion? What is it that you are claiming?
3. Why do you think that?
4. What are the arguments pro and con?
5. What assumptions must we make to accept that conclusion?
6. What is your basis for saying that?

Annotation Scheme and Task with Example:

Argument:

From elementary school all the way to now, I have always done poor academically. I was always wondering if I have an intellectual disorder since I just couldn't grasp concepts as well as the other students.

Argument Structure Diagnosis:

- Highlight core claim, minor claim, premise (span evaluation)
 - Core Claim Span: The main assertion or central argument the author is making—the primary point they want you to accept or consider.
 - Minor Claim Span: A subsidiary or supporting assertion that serves the core claim but isn't the main focus of the argument.
- Example:
 - Core claim: if I have an intellectual disorder
 - Minor Claim: From elementary school all the way to now, I have always done poor academically
 - Premise: I just couldn't grasp concepts as well as the other students

Paraphrasing Initial Understanding:

- Paraphrase the core claim into the most simple sentence.
- Eg: If the author has an intellectual disorder, then it affects academic performance

Inductive vs Deductive Reasoning:

- Select between: **Dropdown: [Inductive, Deductive, None of the above]**
- *Classify whether the argument moves from specific to general (inductive), general to specific (deductive), or doesn't fit either pattern.*
- *Eg: Inductive*

Implicit Assumption:

- Identify if the argument contains the necessary components to make that conclusion (Yes/No) (write short keywords).
- Dropdown: [Yes=Complete, No=Missing Components]
- *Decide whether the argument has all the logical pieces needed for its conclusion. If not, write what is missing as keywords.*
- Eg: Yes

3. Inference

Facione Theory: To identify and secure elements needed to draw reasonable conclusions; to form conjecture and hypothesis; to consider relevant information and to reduce the consequences flowing from data, statement, principle, evidence, judgement, belief, opinion or other form of representation.

1. Given what we know so far, what conclusion can we draw?
2. What are the consequences of doing things that way?
3. What does the alternative can be implied to the conclusion?
4. What additional information do we need to resolve this question?

Annotation Scheme and Task with Example:

Argument:

From elementary school all the way to now, I have always done poor academically. I was always wondering if I have an intellectual disorder since I just couldn't grasp concepts as well as the other students.

Consequence Prediction:

- **If accepting this argument would lead to positive or negative outcomes:**
- **Positive consequences:** Write short keywords or phrases (maximum 2).
- **Negative consequences:** Write short keywords or phrases (maximum 2).
- Eg: Positive: Self-reflection, seeking understanding, long-term awareness, Negative: Poor academic performance, learning difficulties, comparative struggles

Primary Domain Affected:

- **Select one from the dropdown:**[Personal/Physical behaviour, Interpersonal relationships, Social/Cultural norms, Organisational practices, Government/Policy/Law, Economic systems]
- *Pick which area of life would be most impacted if this argument were accepted.*
- Eg: Personal behaviour

Alternative Possibility Recognition:

- Complete one alternative interpretation with 3 keywords maximum:
 - "...this only applies when _____, _____, _____"
 - "...other factors are more important like _____, _____, _____"
 - "...this misunderstands _____, _____, _____"
- **Buttons: [applies when, other factors, misunderstands]**
- *Choose one way to challenge or limit the argument and provide up to 3 keywords.*
- Eg: "...other factors are more important like working hard, teacher ability, and curriculum"

4. Evaluation

Facione Theory: To assess the credibility of statements or other representations that are accounts or descriptions of a person's perceptions, experiences, citations, judgments, or opinions, and to assess the logical strength of the actual or intended inferential relationships among statements, descriptions, questions, or other forms of representation.

- How credible is that claim?
- How do you think we can trust this person claims?
- How strong are those argument
- Do we have fact (necessary information) right?
- How confident we can be in our conclusion, given what we now know?

Annotation Scheme and Task with Example:

Argument:

From elementary school all the way to now, I have always done poor academically. I was always wondering if I have an intellectual disorder since I just couldn't grasp concepts as well as the other students.

Inference:

- How strongly does the premise support the core claim? (1-3 rating)
- Rating: [1=Weak, 2=Moderate, 3=Strong]
- *Rate how well the evidence backs up the main conclusion.*
- Eg: [2]

Assessing Credibility:

- Select the credibility factors: **Checkboxes: [Personal bias, Hearsay/secondhand, Emotional reasoning, Limited expertise, None identified]**
- *Check any factors that might make this argument less reliable or trustworthy.*
- Eg: Personal bias

Assessing Logical Strength:

- Select one from the dropdown list: **[No logical fallacy, False dilemma, Hasty generalisation, Ad hominem, Strawman, Circular reasoning, False cause]**
- *Identify if the argument contains a specific type of flawed reasoning.*
- Eg: No logical fallacy

Can the Claim Can/Cannot Be Trusted:

- Select from **Checkboxes: [Evidence-based, Logical consistency, Common knowledge, Cannot be trusted]**
- *Choose the main reason why this argument should or should not be believed.*

Dropdown Explanations:

- **Inference Score:**
 - **Score 3 (High Support):**
 - Premises directly and logically lead to the core claim (no major assumption required)
 - Evidence is specific, relevant, and sufficient
 - Logical chain is clear with minimal inferential leaps
 - **Score 2 (Moderate Support):**
 - Premises provide some support but with limitations (Requires one significant assumption)
 - Evidence is relevant but incomplete or requires additional assumptions
 - **Score 1 (Weak Support):**
 - Premises provide minimal or questionable support for claim
 - Evidence is irrelevant, insufficient, or anecdotal
 - Major logical gaps or unsupported inferential leaps
- **Credibility Factors:** What might make the argument unreliable (bias, poor sources, etc.)
 - **Personal bias** - The arguer has personal interests that might influence their position (financial stake, emotional investment).
 - **Hearsay/secondhand** - Information comes from rumors, gossip, or unverified sources rather than direct evidence.
 - **Emotional reasoning** - Argument relies heavily on feelings rather than facts or logic.
 - **Limited expertise** - The person lacks relevant knowledge, experience, or qualifications in the topic area.
 - **None identified** - No obvious credibility issues detected in the argument.
- **Logical Strength:** Whether the argument uses faulty reasoning patterns
 - **No logical fallacy** - The argument uses sound reasoning without flawed logic.
 - **False dilemma** - Presents only two options when more exist ("You're either with us or against us").
 - **Hasty generalisation** - Makes broad conclusions from limited examples ("I met two rude people from that city, so everyone there is rude").
 - **Ad hominem** - Attacks the person making the argument instead of addressing the argument itself ("Don't listen to him, he's a dropout").
 - **Strawman** - Misrepresents someone's position to make it easier to attack ("Environmentalists want us all to live in caves").
 - **Circular reasoning** - Uses the conclusion as evidence for itself ("The Bible is true because the Bible says so").
 - **False cause** - Assumes one thing caused another without proof or wrong causation ("I wore my lucky shirt and won, so the shirt caused my victory").

- **Trustworthiness:** The main reason to believe or doubt this argument
 - **Evidence-based** - The argument provides data, studies, or verifiable facts to support its claims.
 - **Logical consistency** - The argument follows clear reasoning without contradictions or logical errors.
 - **Common knowledge** - The claim relies on widely accepted facts or general understanding.
 - **Cannot be trusted** - The argument has significant flaws, bias, or unreliable information that undermines its credibility.

Argument: "Here in the UK schools have to wear a shirt, trousers and a blazer, it is uncomfortable of course. But it is better. But when it comes to bullying, people getting bullied because of what they wear or if they are poor is such a big thing."

Core Claim: "it is better"

Premise: "bullying because of what they wear or if they are poor is such a big thing"

Score: 2 (Moderate support)

Argument: "This isn't some policy where every baby with a severe mental issue MUST be euthanized. This is giving the option to parents to euthanize their baby. You can't extend this to euthanizing adults, because there's no comparable circumstance where an individual has control over an adult."

Core Claim: "You can't extend this to euthanizing adults"

Premise: "there's no comparable circumstance where an individual has control over an adult"

Score: 2 (Moderate support)

Argument: "I believe that ACAB just like I believe that all Nazi guards are bastards. Good cops don't exist because you can't fulfill your orders as a cop and not be a bastard. (e.g. Arresting for possession should not be thing)."

Core Claim: "Good cops don't exist",

Premise: "you can't fulfill your orders as a cop and not be a bastard. (e.g. Arresting for possession should not be thing)",

Score: 1

5. Explanation

Facione Theory: To state and to justify that reasoning in terms of the evidential, conceptual, methodological, criteriological, and contextual considerations upon which one's results were based; and to present one's reasoning in the form of cogent arguments.

- What were the specific findings/results of the investigation (Analysis >> Claim)?
- Please tell us how you conducted that analysis. (Analysis)
- How did you come to that interpretation? (Interpretation)
- Please take us through your reasoning one more time. (overall thinking)
- Why do you think that was the right answer/was the solution? (Inference)
- How would you explain why this particular decision was made?

Annotation Scheme and Task with Example:

Argument:

From elementary school all the way to now, I have always done poor academically. I was always wondering if I have an intellectual disorder since I just couldn't grasp concepts as well as the other students.

Target Explanation:

I identified [Target] as the primary target because [dropdown list]

[It is explicitly stated as the conclusion (e.g., "therefore/thus/so"). |

It is repeated or foregrounded through emphasis and evidence selection. |

It is the core causal claim (X causes Y) that drives the argument's implication. |

It is stated early and framed with clear evaluative language (good/bad/should).]

Example: I identified [intellectual disorder] as the primary target because [It is the core causal claim (X causes Y) that drives the argument's implication.]

Argument Structure: *"Review your selected argument components:" Annotator can use exact span or paraphrase version to fill in the blank*

- (Span1) Core Claim: [Previous selection], (Span2) Minor Claim: [Previous selection], (Span3) Premise: [Previous selection]

Explanation Reasoning structure (Select one and fill in the blank)

- Uses specific cases, stories, or instances to support claims → "Particular case [Span1] → supports broader principle [Span2] → therefore [Span3]"
- Justifies action by their positive or negative outcomes → "If we do [Span1] → then [Span2] → therefore [Span3]"
- Compare current citation to similar known situation → *"Situation A [Span1] is like*

situation B [Span2] → what is true of B applies to A [Span3]"

- Use indicator to suggest underlying condition → "Observable evidence [Span1] → suggests pattern [Span2] → indicates condition [Span3]"
- Appeal to expert knowledge or credible source → "Source [Span1] states [Span2] → therefore [Span3] is credible"
- None of the others

Example:

Argument Structure Shown:

- Core claim: if I have an intellectual disorder
- Minor Claim: From elementary school all the way to now, I have always done poor academically
- Premise: I just couldn't grasp concepts as well as the other students

Shown Choice:

- Use indicator to suggest underlying condition → "Observable evidence [I just couldn't grasp concepts as well as the other students] → suggests pattern [From elementary school all the way to now, I have always done poor academically] → indicates condition [if I have an intellectual disorder]"

Inductive vs. Deductive:

- Show the previously selected label: **Argument** → **[Label]**
- Paraphrase of terminology is acceptable
Select a span to indicate either:
 - **General principle [Term X] → Specific conclusion [Term Y],** or
 - **Specific instance/principle [Term X] → General principle [Term Y]**

Example: *Inductive*

- Specific instance/principle [Bad performance] → General principle [if I have an intellectual disorder]

Evidential Justification:

"I made this inference based on the following evidence from the text, in my own words:"

- **Good:** Short text (keywords / brief phrases)
- **Bad:** Short text (keywords / brief phrases)

Example: Positive: Seek support, Negative: Self-Stigmatization

Positive: Author can seek support after self realization

Bad: Self-stigma of learning disorder

Explaining Why the Claim Can/Cannot Be Trusted:

Show the previous checkbox: [Result of checkboxes]

- Then explain in short keywords based on the dropdown selection.
- Eg: Common knowledge that intellectual disorders cause poor academic performance

Fallacy Justification

No logical fallacy: The argument uses sound reasoning and does not contain a clear flaw.

False dilemma (either–or): The argument presents only two options when more possibilities exist.

Template: Option A **[Term1]** vs. Option B **[Term2]** (missing other options)

Hasty generalization: The argument draws a broad conclusion from too few or unrepresentative examples.

Template: **[Term 1]** is too broad / the evidence is insufficient or not representative

Ad hominem: The argument attacks the person (or their character) instead of addressing the claim or reasons.

Template: Attacks **[Person/Group]** → therefore rejects **[Claim]** (without addressing the claim)

Strawman: The argument misrepresents someone's position to make it easier to criticize.

Template: **[Term1 / Original position]** is distorted into **[Term2 / Weaker position]**

Circular reasoning (begging the question): The conclusion is used as its own support (restated as evidence).

Template: **[Term1 / Conclusion]** is used as evidence for **[Term1 / Conclusion]**

False cause: The argument assumes a causal relationship without sufficient proof (or confuses correlation with causation).

Template: **[Term1]** does not necessarily cause **[Term2]** (causal link is unsupported)

Example: No logical fallacy

6. Self Regulation

Facione Theory: Facione defines **self-regulation** as the ability to *monitor, question, and revise one's own reasoning and judgments*, especially by checking for errors, bias, or unwarranted confidence

Sub-skills:

Self-examination → *"Was my previous judgment influenced by fear, bias, or heuristic shortcuts?"*

Self-correction → *"Given this reflection, do I want to revise my earlier decision?"*

- Our position on this issue is still too vague, can we be more precise?
- How good was our methodology, and how well did we follow it?
- Is there a way we can reconcile these apparently conflicting conclusions?
- How good is our evidence?
- Ok, before we commit, what are we missing?
- I'm finding some of our definition a little confusing, can we revisit what we mean by certain things before making any final decision?

Main Question:

In your previous response, you may have been influenced by an immediate emotional reaction (fear, anxiety). Now that you have more time to reflect systematically, would you like to reconsider your decision?

Pre-Regulation Display:

Show complete previous annotation history:

- **Interpretation Phase Result:** [Previous selection]
- **Analysis Phase Result:** [Previous selection]
- **Inference Phase Result:** [Previous selection]
- **Explanation Phase Result:** [Previous justification]

Annotation Scheme and Task with Example:

Argument:

From elementary school all the way to now, I have always done poor academically. I was always wondering if I have an intellectual disorder since I just couldn't grasp concepts as well as the other students.

1A. Bias Detection (up to 2):

"Looking at your previous responses, were any influenced by:" **Checkboxes:** ☐ Fear/anxiety ☐ First impression ☐ Emotional reaction ☐ Pattern matching ☐ Insufficient self-checking ☐ None detected

Example: ☐ First impression, ☐ Pattern matching

1B. Heuristic Shortcut Detection (up to 2):

"Did you use any mental shortcuts that might have affected accuracy?" **Checkboxes:** ☐ Went with gut feeling ☐ Used familiar pattern ☐ Followed surface cues ☐ None detected -**Example:** ☐ Went with gut feeling

1C. Error Detection:

"Do you now notice any potential errors in your previous reasoning?" Short text: *"Describe what you notice (or 'None detected')"*

Example:

2A. Change Decision: *"Do you want to change any of your previous responses?"* **Simple choice:** Yes | No

Example: Yes

2B. Revision Decision (if Yes) *"Which phases would you like to revise?"* Checkboxes: ☐ Interpretation ☐ Analysis ☐ Evaluation ☐ Inference ☐ Explanation

Example: ☐ Analysis

2C. Revision Reason: *"Why are you choosing to revise?"* Template: *"I am revising because [SELF-EXAMINATION INSIGHT]"*

Example: I realise I was influenced by the emotional tone and used pattern matching to quickly label elements as negative without fully considering the complexity of self-questioning as a cognitive strength

2D. Revision Type: *"How significant is your revision?"* Dropdown: minor-adjustment | major-change | complete-revision | no-change

Example: minor-adjustment

1A. Bias Detection:

"Looking at your previous responses, were any influenced by:"

- ☐ **Fear/anxiety** - Worried about negative consequences or being wrong influenced my judgment
- ☐ **First impression** - My immediate reaction when reading the argument stuck throughout analysis
- ☐ **Emotional reaction** - Personal feelings (anger, sympathy, disgust) affected my reasoning
- ☐ **Pattern matching** - Automatically categorized based on similarity to familiar arguments
- ☐ **Insufficient self-checking** - Made quick annotation decisions without double-checking or questioning my choices
- ☐ **None detected** - No emotional or cognitive biases influenced my responses

1B. Heuristic Shortcut Detection:

"Did you use any mental shortcuts that might have affected accuracy?"

- ☐ **Went with gut feeling** - Trusted my instinct instead of systematic step-by-step analysis or avoided complex analysis to reduce mental effort
- ☐ **Used familiar pattern** - Applied a known framework from previous annotation. without checking if it truly fits
- ☐ **Followed surface cues** - Based judgment on obvious keywords/phrases rather than deep meaning (For example, discourse indicator)
- ☐ **None detected** - Used thorough, systematic analysis throughout

7. Socratic Question Annotation

Based on your critical thinking analysis across the six phases, generate up to three Socratic questions.

Argument: "Because it's one shared nation with values and culture. That is the only basis of a nation that exists. Only the shared culture and values.",

1. Do you think political scientists would agree with your definition of nations?

Target: Challenges the definition against academic understanding

Source: "initialUnderstandingTarget": "nation basis",

2. Can you think of any successful nations that were formed primarily through economic unions, geographic necessity, or political agreements rather than shared culture?

Target: Challenges the "only" claim with alternative cases

Source: "alternativeType": "other-factors",

"alternativeKeywords": "citizen well-being, economy",

3. What other factors would contribute to the sustainability of a nation, such as citizen well-being and the economy?

Target: Challenges comprehensiveness of the explanation

Source: "alternativeType": "other-factors",

"alternativeKeywords": "citizen well-being, economy",

Argument: "Here in the UK schools have to wear a shirt, trousers and a blazer, it is uncomfortable of course. But it is better. But when it comes to bullying, people getting bullied because of what they wear or if they are poor is such a big thing."

1. Do you have any concrete evidence to support the claim that school uniforms are better, rather than just saying they prevent bullying?

Target: Challenges unsupported evaluative claim

Source: "coreClaim": "it is better",

2. What could be the positive consequences for society and the community of not wearing school uniforms, including individuality and self-expression?

Target: Identifies what's gained by NOT implementing the policy

Source: "alternativeKeywords": "individuality, self-expression",

Argument: "I believe that ACAB just like I believe that all Nazi guards are bastards. Good cops don't exist because you can't fulfill your orders as a cop and not be a bastard. (e.g. Arresting for possession should not be thing)."

1. Don't you think it is a hasty generalization to conclude that all cops are bad?

Target: Challenges the universal generalization from limited examples

Source: "paraphrasingUnderstanding": "All cops are bad.",

2. Do you think you understand police activity correctly to make such a conclusion?

Target: Questions whether author has adequate understanding

Source: "alternativeKeywords": "police activity, hardworking officers",

3. Is it possible that you are exercising personal bias to reach the conclusion that all cops are bad?

Target: Suggests personal bias may influence conclusion.

Source: "trustExplanation": "Cannot be trusted because the evidence feels extremely weak and biased."